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RESEARCH-ARTICLE

Leveraging WebTraceSense for User Interaction Log Analysis: A Case Study on a Visual Data Analysis Tool for the Visualization of User Interactions Logs

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Abstract

The current surge in the development of web applications highlights the necessity of incorporating user-specific preferences into the design process. An innovative approach to improving these applications involves the analysis of interaction data recorded by browsers, such as the number of mouse clicks and keystrokes. The data thus obtained provides valuable insight into user behavior, enabling effective personalization of web applications. The WebTraceSense project proposes the development of a web platform designed to facilitate the customization of the visualization of interaction data from websites. The platform will include a dynamic visualization component, which will support the identification of user behaviors, and a DevOps cycle, which will help streamline software cycle processes. This article presents a case study for the examination of user interaction logs from a visual data analysis tool, utilizing the functionalities of the WebTraceSense platform to facilitate the identification of behavioral trace patterns.

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CCS Concepts

• **Human-centered computing;**; • **Accessibility;**; • **Accessibility systems and tools**;

Keywords

Accessibility, User Log, Visualization

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1 INTRODUCTION

Ecommerce websites, web games, and other digital platforms stand to gain significantly from personalization. By leveraging user data to tailor content and experiences, these platforms can improve user satisfaction and engagement. For instance, personalized recommendations in ecommerce can increase sales and customer loyalty by providing users with products that match their preferences [1].

Market statistics indicate that platforms utilizing personalized content see higher user retention rates and better overall performance metrics [2]. In today's rapidly evolving gaming landscape, the personalization of web games has become crucial for providing an engaging and tailored experience to players. In web games, personalization can adapt the game difficulty and narrative to suit individual player styles, enhancing their enjoyment and prolonging their engagement with the game. By tailoring the gaming experience to individual players, developers can enhance user engagement and satisfaction. The concept of gamification has been increasingly utilized in various fields, including marketing, education, and business, to engage and motivate users [3]. In the context of web games, gamification plays a pivotal role in creating immersive and interactive experiences for players. By incorporating game-like elements such as points, badges, and leaderboards, web games can capture and maintain the attention of users, ultimately enhancing their overall experience. Furthermore, the utilization of gamification techniques can facilitate player retention and encourage repeated gameplay, thereby contributing to the success of web games in the digital landscape [4]. Other areas such as web platforms for educational contexts also benefit from personalization, for example in the adaptation of the interface tailored to users with intellectual disabilities [5].

Digital phenotyping is an approach that utilizes data collected from digital devices to assess and understand user behaviors and traits. This approach can provide deep insights into user interactions and preferences. Task fingerprinting, a related concept, refers to the unique patterns and behaviors exhibited by users when interacting with digital tasks. By analyzing these patterns, developers can create highly personalized experiences tailored to the specific behaviors and preferences of users [6]. These techniques are invaluable in enhancing the personalization of web applications, including games, by providing a granular understanding of user engagement. In order to identify behavior patterns and inform personalization strategies, it is crucial to create a web platform for visualizing user log interaction data. Such a platform can provide developers with intuitive tools to analyze interaction data, such as mouse clicks and keystrokes, enabling them to identify trends and preferences. By visualizing this data, developers can make informed decisions about game design and content delivery, which in turn enhances the user experience [7]. The WebTraceSense project aims to build such a platform, providing a comprehensive solution for capturing, analyzing, and visualizing interaction data from websites [8]. This platform will help developers to understand user behaviors more deeply, facilitating the creation of more engaging and personalized web applications.

The objective of this article is to present a case study which demonstrates the utilization of the WebTraceSense platform for the visualization of user log interactions within the context of a data analysis scenario. The exploration of WebTraceSense in customizing web application interaction data has the potential to yield significant outcomes for the web development industry. With a deeper understanding of user behaviors and preferences, developers can optimize app design, functionality, and content to align with the specific needs and interests of their target audience. Furthermore, the utilization of personalized data enables web applications to present tailored experiences, thereby enhancing user engagement and satisfaction. The WebTraceSense web platform is designed to facilitate the generation of visualizations of interaction data recorded by browsers, such as mouse clicks and keystrokes. This will contribute to the advancement of personalized web experiences.

2 BACKGROUND

2.1 Digital phenotyping and Task fingerprinting

User log analysis can be applied in a multitude of scenarios. An article discusses the use of AI and related technologies to merge traditional and digital biomarkers for the purpose of improving patient diagnosis and treatment, also designated as digital phenotyping [9]. It highlights the potential of "point-of-care" (POC) diagnostics, which are medical tests performed at or near the site of patient care. The POC platform enables rapid and real-time feedback to physicians, thereby enhancing the quality of care by providing immediate information about the patient's condition. This approach is particularly valuable in healthcare settings where quick decision-making is critical. A study presented on the webpage examines the potential of utilizing touchscreen typing patterns to remotely detect depressive tendencies [10]. The study introduces a machine learning-based method that differentiates between individuals with depressive tendencies and healthy controls by analyzing keystroke timing sequences and typing metadata collected from smartphone interactions. The method demonstrated high accuracy, with an AUC of 0.89, and a significant correlation with self-reported PHQ-9 scores, suggesting its effectiveness in high-frequency monitoring of depressive tendencies in daily life.

Furthermore, user logs can be analyzed in order to infer behavioral traces in e-commerce platforms, or even in digital work scenarios. A study found that the performance of crowdsourcing microtasks can be evaluated not only in terms of the outcomes of the microtasks (e.g. accuracy), but also by analyzing how the work is performed (e.g. user log interactions). This can be a useful indicator of work quality [11]. The data is derived from simple logs generated by JavaScript libraries, which record details of user interactions with the browser, such as click events or key presses. The log obtained can be used to identify the user behavior, such as whether the user is a lazy or diligent worker. Furthermore, the identification process, known as task fingerprinting, was subjected to testing on classification, generation and comprehension microtasks. Subsequently, models were generated based on machine learning techniques, resulting in satisfactory prediction outcomes. It can be demonstrated that users are able to extract useful information from low-level Web log data, provided that the data mining features facilitate these activities [12]. The research indicates that higher order knowledge discoveries are facilitated by tool assistance or supported workflows. Both the perceived utility and the greater quantity of actionable knowledge discoveries offset the perceived difficulty of the aided processes. Other articles describe an interactive front-end visualization tool for the specific analysis of natural language processing [13, 14].

2.2 Game Analytics based on logging user interactions

User log analysis can also be applied in game scenarios contexts, where understanding player behavior and preferences can lead to more engaging and personalized gaming experiences. Related works have demonstrated the potential of leveraging user interaction data to enhance game design and user experience. Also named game analytics, Gomez and colleagues [15] conducted a systematic literature review in this field and stated that it has gained significant attention in recent years, as game developers strive to create more immersive and tailored gaming environments. One focus in game analytics is the cognitive skills, which include the core abilities used to think, read, learn, remember, reason, and pay attention, are crucial for effective information processing. In the context of web interaction analysis, understanding how users process new information and distribute it into appropriate areas of the brain can significantly enhance the personalization of web applications. Developing these cognitive skills can lead to faster and more efficient processing of information, thereby improving user experience. Furthermore, the evaluation of daily physical tasks can provide insights into user interaction patterns and preferences. For instance, Rodríguez-de Pablo [16] employed a set of games to provide a fast, quantitative, and automatic evaluation of arm movement function, demonstrating how physiological assessments can be integrated into web platforms.

The analysis of web interaction data involves a range of methodologies to bridge the gap between user behavior and application assessment. Gómez and colleagues [15] grouped five primary groups of methods: Descriptive Statistics, Machine Learning, Knowledge Inference, Deep Learning, and Sequence Mining. Descriptive Statistics includes mathematical computations, evaluations, and visual representations. Numerous studies have implemented summary metrics, such as averages and variances, correlations, and visual data depictions to interpret web interaction information. These techniques provide a basic understanding of data patterns and distributions, facilitating preliminary insights into user behavior. Machine Learning, a subset of AI, encompasses methods that allow systems to learn and improve from past data. The two primary categories of machine learning techniques are supervised learning and unsupervised learning. Supervised learning methods, such as regression, and unsupervised techniques, like k-means clustering and principal component analysis (PCA), have been widely used [17]. Deep Learning is an AI function that mimics the human brain's data processing to create patterns for decision-making. Deep learning models are particularly adept at identifying complex patterns and making predictions based on large datasets, which is highly beneficial in understanding the nuances of user behaviors [18]. Knowledge mining involves deriving new knowledge from existing data using rules and constraints. Common techniques include Bayesian networks and fuzzy cognitive maps. Finally, sequence mining is a method aimed to uncover valuable knowledge from sequential data. Gomez and colleagues [19] applied sequence mining to identify patterns and errors in user interactions, transforming raw data into meaningful sequences that can support a more accurate interpretation and analysis.

By employing cognitive and soft skills analysis, as well as physiological assessments, as previously documented in other research works, it is possible to provide users with a highly personalized and dynamic experience when utilizing web applications.

3 WEBTRACESENSE PLATFORM FOR THE VISUALIZATION OF USER LOGS

The WebTraceSense platform aims to provide a comprehensive solution for analyzing web interaction data [8]. The project involves the development of a backend using .NET Core to create a Restful API that manages resources associated with the analysis of interaction data. This backend is complemented by the establishment of a database designed to store interaction log files in JSON format, ensuring efficient data management and retrieval. On the frontend, the development focuses on creating a web application using React, to facilitate the visualization of interaction data. This involves not only the creation and editing of HTML/CSS pages but also the implementation of a step-by-step web form, commonly referred to as a "Wizard," which helps users customize the visualization of their web interaction records. Additionally, registration and login pages are developed, adhering to web usability heuristics to ensure a smooth and intuitive user experience. A significant aspect of the WebTraceSense platform is the development of visualization modules using JavaScript libraries such as D3.js. These modules are essential for illustrating interaction data in a dynamic and interactive manner, providing users with clear and insightful visual representations of their data.

To ensure that the platform operates efficiently in a production environment, a robust DevOps and automation process is implemented. This includes the deployment of backend and frontend components into production mode, conducting thorough integration tests to validate both components, and creating containerized solutions using Docker. These steps are crucial for maintaining the reliability and scalability of the platform. An overview of the WebTraceSense Project (see Figure 1), begins with the requirement that the researcher or any individual interested in analyzing user interaction logs must have a website from which these logs can be gathered (Label 1 in Figure 1).

The user interaction logs are then exported to a JSON file, capturing key metrics such as click data, key presses, and user 's session latency (Label 2 in Figure 1). This includes the deployment of backend and frontend components into production mode, conducting thorough integration tests to validate both components, and creating containerized solutions using Docker. These steps are crucial for maintaining the reliability and scalability of the platform. An overview of the WebTraceSense Project (see Figure 1), begins with the requirement that the researcher or any individual interested in analyzing user interaction logs must have a website from which these logs can be gathered (Label 1 in Figure 1). The user interaction logs are then exported to a JSON file, capturing key metrics such as click data, key presses, and user 's session latency (Label 2 in Figure 1).

User authentication is facilitated through the implementation of the OAuth protocol, allowing users to log in using a third-party provider. Once authenticated on an external platform, users are redirected to the WebTraceSense frontend (Labels 3 and 4 in Figure 1). On the main webpage, users are presented with CRUD (Create, Read, Update, Delete) options for managing the visualizations of user logs. The creation process is guided by a wizard form (Label 5 in Figure 1), which involves multiple steps, including the uploading of interaction log files (Label 6 in Figure 1) and the selection of metrics

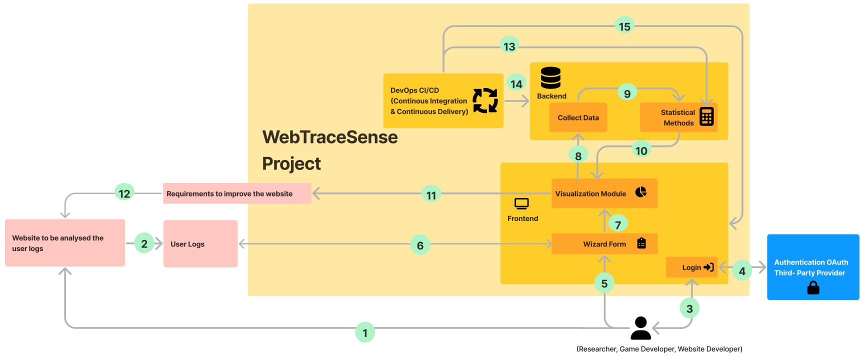

Figure 1: Overview of the WebTraceSense Project.

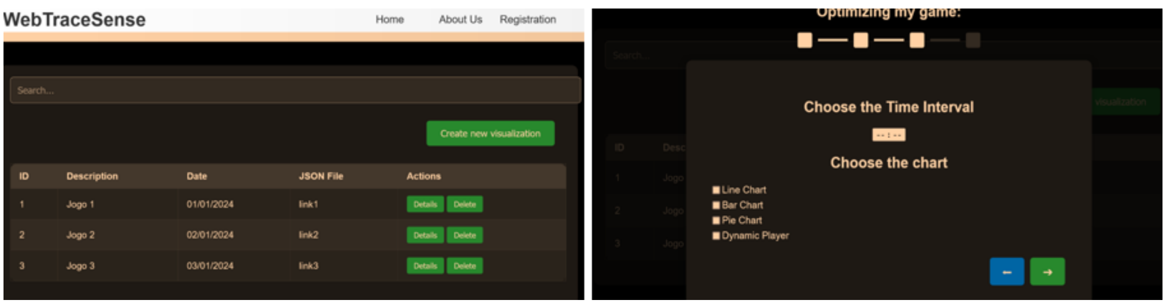

Figure 2: CRUD page (left) and wizard form (right) in the WebTraceSense frontend.

to be analyzed, such as mouse data, key presses, and temporal intervals, as well as the types of visualization charts to be generated.

Once the visualization request is created, it is processed in the Visualization module, which utilizes D3.js for generating dynamic data-driven graphics (Label 7 in Figure 1). The backend collects data on the visualization metrics used, and this data is then subjected to statistical analysis to enhance the generated visualizations (Labels 9 and 10 in Figure 1). The resulting visualizations support insights, helping the definition of requirements for researchers, game developers, or web developers to improve their web platforms (Labels 11 and 12 in Figure 1). This iterative process allows for continuous refinement of the web platform, generating new user interaction logs and performing subsequent analyses. Throughout this process, the WebTraceSense project incorporates a DevOps cycle to automate the integration of statistical methods (Label 13 in Figure 1) into data-driven processes. This cycle also facilitates the transition of backend (Label 14 in Figure 1) and frontend (Label 15 in Figure 1) components from the development to the production environment, employing unit and integration tests to ensure a seamless and efficient operation.

Figure 2 presents two principal web pages of the WebTraceSense frontend. The initial page illustrates the principal CRUD options, which facilitate the administration of the visualizations generated from the user interactions log. The subsequent page depicts a four-step wizard form, through which researchers, game developers and website programmers can upload the user log file and generate a tailored visualization to gather insights on the user experience. The initial step in the wizard form provides a brief tutorial. The second step presents the option to upload the JSON file of the user log. The third step (shown in Figure 2 – right image) allows the user to specify the time interval and the type of charts they would like to produce. The fourth step presents a preview of the generated visualization. The generated visualization can have multiple options, and the main types of charts are the line chart, bar chart, pie chart and the dynamic player (see Figure 3). The Dynamic Player differs from the other visualization options in that it allows a more realistic representation of user behavior by drawing the corresponding interaction (i.e. mouse or key press) in a set of screenshots of the interface where the user logged in. Furthermore, it also includes the option of representing a heatmap where the user clicked more often.

4 CASE STUDY FOR THE EXAMINATION OF USER INTERACTION LOGS FROM A VISUAL DATA ANALYSIS TOOL

To demonstrate the capabilities of the WebTraceSense system, we analyzed a case study that collected user interaction logs from a visual data analysis tool, named Lumos [20]. The user interaction logs were made available in a public dataset designed to enhance user awareness of their analytic behavior during data exploration. The tool records the user's interaction history and provides immediate feedback by displaying interaction traces. The aforementioned interaction traces serve to highlight the specific parts of the dataset that users have focused on, thereby facilitating an opportunity for users to reflect on any potential biases or overemphasis that may have been exhibited during the course of their analysis. Lumos incorporates in-situ visual feedback through the use of color-coding, whereby elements are colored based on the frequency of user interactions. This enables users to compare their own behavior with that of the underlying data distribution, thus facilitating the identification of potential biases and the refinement of their analysis strategies.

Lumos' principal contributions are the capture of interaction data (e.g. hovers and clicks) and the creation of awareness through a combination of in-situ and ex-situ visual feedback mechanisms. A user study comprising 24 participants demonstrated that Lumos enhances users' awareness of their data exploration practices, facilitating self-reflection on unconscious biases and encouraging more informed decision-making. The tool also supports customizable target distributions, enabling users to compare their behavior against predefined goals, such as achieving diversity in university admissions. The Lumos study employed a dataset comprising 709 films, with attributes including production budget, worldwide gross, running time, IMDB rating, Rotten Tomatoes rating, release year, content rating, and genre. Participants in the study investigated these attributes with a view to recommending movie characteristics for a production company, analyzing the impact of various attributes, such as ratings and budgets, on decisions regarding the types of movies that should be produced next.

In comparison to alternative visualization options, the Dynamic Player offers a more realistic representation of user behavior. The software captures user interactions, such as mouse clicks or key presses, and superimposes them on screenshots of the user's login interface, thereby highlighting the areas where users interacted the most. Figure 3 illustrates the potential interactions that could be registered in the Lumos UI, showcasing the heatmap functionality.

Furthermore, Figure 3 includes two line charts comparing user interaction logs from two versions of a web application, one with the awareness condition and one without, as referenced in [20]. The chart effectively demonstrates the timing between user interactions. In the awareness version (on the left), there is a clear reduction in the time between interactions, indicating a smoother and more intuitive experience. In contrast, the non-personalized version (on the right) shows longer intervals, suggesting a more difficult interface that requires more time to navigate. This visualization highlights the potential benefits of personalized interfaces, supporting the idea that tailored systems improve user engagement and task performance by adapting to individual preferences.

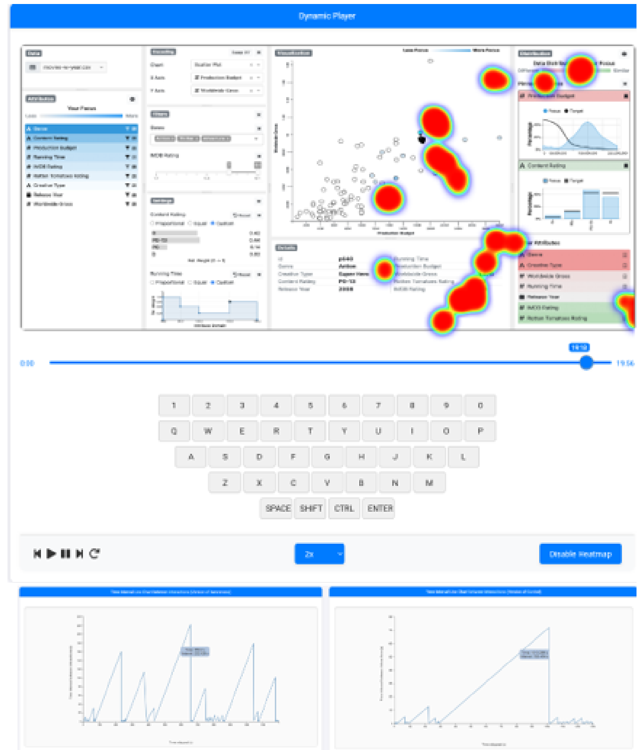

Figure 3: Dynamic player with the heatmap option for the visualization of a user log in an example of Lumos UI (top). A line chart showing one user log for the version with awareness condition (bottom left) and another for the version of control condition (bottom right). The user log is from the dataset used in the Lumos UI study [20].

5 DISCUSSION

The WebTraceSense project establishes a comprehensive framework for analyzing user logs on web platforms, defining the necessary components and their communication flow to facilitate this analysis. This project outlines a series of activities that begin with the collection of user interaction logs and culminate in the seamless generation of visualizations that provide insights into user behaviors. The proposed implementation of this project consists of an end-to-end system comprising a frontend, a backend, visualization modules, a statistical/AI module, and a DevOps cycle that refines these modules. One of the key strengths of the WebTraceSense platform is its user-friendly approach to generating visualizations. Users only need to define a few parameters, such as the time interval, type of chart, and upload the respective logs. This simplicity ensures that even those with limited technical expertise can leverage the platform to gain valuable insights. Once the visualizations are generated, statistical methods or machine learning techniques can be applied to refine these visualizations and identify patterns in user behavior. Additionally, the DevOps cycle integrated into the WebTraceSense platform ensures continuous improvement and scalability. By automating the deployment and testing of both backend and frontend components, the platform maintains high reliability and performance.

This automated process also facilitates the integration of new features and updates, ensuring that the platform evolves in response to user needs and emerging technological advancements. The case study, which was conducted on a visual data analysis tool, helped to validate the potential for the usage of WebTraceSense for the visualization of behavioral traces. The visualization of behavioral traces is well-documented in scientific literature. For instance, the work of Rzeszotarski and Kittur [21] demonstrated the potential of visualizing interaction data to uncover user behaviors. However, despite these advancements, there remains a significant gap in tools that enable researchers, game developers, and website programmers to seamlessly identify behavioral traces through ad-hoc visualizations tailored to their specific needs. WebTraceSense addresses this gap by providing a robust platform that supports the creation of customized visualizations. This automated process also facilitates the integration of new features and updates, ensuring that the platform evolves in response to user needs and emerging technological advancements. The case study, which was conducted on a visual data analysis tool, helped to validate the potential for the usage of WebTraceSense for the visualization of behavioral traces. The visualization of behavioral traces is well-documented in scientific literature. For instance, the work of Rzeszotarski and Kittur [21] demonstrated the potential of visualizing interaction data to uncover user behaviors. However, despite these advancements, there remains a significant gap in tools that enable researchers, game developers, and website programmers to seamlessly identify behavioral traces through ad-hoc visualizations tailored to their specific needs. WebTraceSense addresses this gap by providing a robust platform that supports the creation of customized visualizations.

The inclusion of advanced statistical and AI modules further enhances the platform's ability to provide deep insights into user behavior. These modules can apply sophisticated techniques to interaction data, revealing nuanced patterns that may not be immediately apparent through basic visualization methods.

6 FINAL REMARKS AND FUTURE WORK

The WebTraceSense platform offers a comprehensive solution for the analysis and visualization of web interaction data, with the objective of optimizing the personalization and efficacy of web-based games and applications. The platform provides a means of administering visualizations generated from user interaction logs. Furthermore, a four-step wizard form allows researchers, game developers, and website programmers to upload user log files and generate tailored visualizations for the purpose of gathering insights into the user experience. The generated visualizations offer a variety of options, including line charts, bar charts, pie charts, and the Dynamic Player. The Dynamic Player is distinguished by its capacity to present a more realistic representation of user behavior, exhibiting corresponding interactions such as mouse clicks or key presses on a series of interface screenshots. Furthermore, the platform offers the option of generating a heatmap, which highlights areas where users have clicked with the greatest frequency. The integration of these visualization tools with robust backend and frontend components, automated DevOps processes and statistical analysis methods provides the WebTraceSense platform with a powerful framework for understanding and improving user interactions. This comprehensive approach not only permits a detailed examination of user behavior, as demonstrated in the conducted case study, but also facilitates the continuous refinement of web platforms, thereby ensuring a more personalized and effective user experience.

It is anticipated that the platform will undergo further refinement in the future, including the development of additional visualizations. Moreover, the dynamic player will be augmented by the generation of logs of the interface based on user interactions, thereby obviating the necessity for screenshots for these visualizations.

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